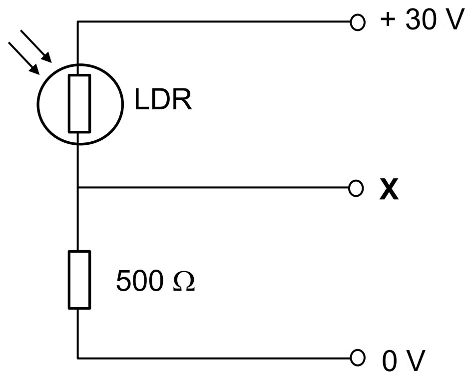
**24.** The light dependent resistor (LDR) and a 500 Ω resistor form a potential divider between voltage lines held at +30 V and 0 V as shown.




The resistance of the LDR is 1000 Ω in the dark but drops to 100 Ω in bright light.

What is the corresponding change in the potential at **X**?

**A** A rise of 25 V

**B** A rise of 15 V

**C** A fall of 15 V

**D** A fall of 25 V

**Answer: B**

In dark, p.d. across LDR $= \dfrac{1000}{1000+500} \times 30 = 20$ V, so potential at X = +10 V

In bright light, p.d. across LDR $= \dfrac{100}{100+500} \times 30 = 5$ V, so potential at X = +25 V

Change in potential $= (+25) - (+10) = +15$ V